

3(a)	1 <i>symbols</i>: normal (allele) = E and EOPD (allele) = e ; 2 <i>parent genotypes + gametes</i>: Ee x Ee and E e E e ; 3 <i>offspring genotypes</i>: EE Ee (Ee) ee ; 4 <i>ratio of offspring phenotypes</i>: 3 normal : 1, EOPD / with disease ;	4
3(b)	any five from: 1 <u>missense</u> mutation ; 2 mRNA forms with different, codon / triplet ; 3 different tRNA, binds / base pairs ; 4 amino acid / primary structure (of, polypeptide / protein / kinase), changed / different ; 5 tertiary / 3D, structure (of, polypeptide / protein / kinase) changed ; 6 (change) may affect / distort, active / catalytic / ATP binding, site ; 7 <u>nonsense</u> mutation ; 8 forms (early) STOP codon ; 9 no, matching tRNA / amino acid added ; 10 shortened / truncated / incomplete, polypeptide ;	5
3(c)	any two from: 1 only need one allele to produce inhibitor ; 2 other proteins cannot be, phosphorylated / activated ; 3 no / less, aerobic respiration / ATP, in (mitochondria of) neurones ; 4 (less ATP so) problem with, dopamine recycling / sodium-potassium pump / establishing resting potential / repolarisation; 5 neurones die ;	2